MATH 345 Skeleton Notes

Unit 3: Beyond linear maps: local linearization and first-order optimization

Section 3.6: Directional derivatives, critical points, and gradient descent

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Directional derivatives

By Linear Approximation, the first-order change in a scalar-valued function near $\mathbf{a}$ is

$$f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) \approx \nabla f(\mathbf{a})^T \mathbf{h}.$$

To study change along a unit direction $\mathbf{v}$, restrict the input to the line

$$\mathbf{c}(t) = \mathbf{a} + t\mathbf{v}.$$

The resulting one-variable rate of change is the directional derivative. The chain rule for one-parameter inputs in Chain Rule for One Independent Variable gives the gradient formula below.

Definition

Let f be a scalar-valued function. Then if $\mathbf{v}$ is a unit vector, the directional derivative of f in the direction of the vector $\mathbf{v}$, denoted $D_{\mathbf{v}} f$, is defined as

$$D_{\mathbf{v}} f(\mathbf{a}) = \lim_{h \to 0} \frac{f(\mathbf{a}+h\mathbf{v}) - f(\mathbf{a})}{h},$$

provided the limit exists.

The directional derivative represents the rate of change of the function f at the point $\mathbf{a}$ in the direction of the unit vector $\mathbf{v}$. For a function $f(x, y)$ of two variables, the directional derivative $D_{\mathbf{v}} f(\mathbf{a})$ is the slope of the tangent line to the curve obtained by intersecting the surface $z = f(x, y)$ with the plane containing both the line passing through $\mathbf{a}$ with direction vector $\mathbf{v}$, and the z axis.

Remark

Partial derivatives are a special case of directional derivatives, i.e., we have

$$\frac{\partial f}{\partial x_i} = D_{\mathbf{e}_i} f.$$

Despite partial derivatives being a special case of directional derivatives, in most circumstances we can compute directional derivatives using partial derivatives, or equivalently, the gradient.

Theorem

Suppose f is a function differentiable at a point $\mathbf{a}$. Then for each unit vector $\mathbf{v}$, $D_{\mathbf{v}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v}$.

Why is this true?

Define the path $\mathbf{c}(t) = \mathbf{a} + t\mathbf{v}$ and the single-variable function $g(t) = f(\mathbf{c}(t)) = f(\mathbf{a} + t\mathbf{v})$. Since $\mathbf{c}(0) = \mathbf{a}$, we have $g(0) = f(\mathbf{a})$. Therefore, the definition of the directional derivative gives

$$\begin{aligned} D_{\mathbf{v}} f(\mathbf{a}) &= \lim_{h \to 0} \frac{f(\mathbf{a} + h\mathbf{v}) - f(\mathbf{a})}{h} \\ &= \lim_{h \to 0} \frac{g(h) - g(0)}{h} \\ &= g'(0). \end{aligned}$$

Write $\mathbf{a} = (a_1, \dots, a_n)$ and $\mathbf{v} = (v_1, \dots, v_n)$. The ith coordinate of $\mathbf{c}(t)$ is $c_i(t) = a_i + tv_i$, so $c_i'(0) = v_i$. Because $g = f \circ \mathbf{c}$, the chain rule (Chain Rule for One Independent Variable) gives

$$\begin{aligned} g'(0) &= \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\mathbf{c}(0))c_i'(0) \\ &= \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\mathbf{a})v_i \\ &= \nabla f(\mathbf{a}) \cdot \mathbf{v}. \end{aligned}$$

Combining the two calculations proves that $D_{\mathbf{v}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v}$.

The theorem greatly simplifies the computations of directional derivatives.

Activity: Alignment with the gradient (U3-LO8, U1-LO1)

Suppose

$$\nabla f(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

Compute $D_{\mathbf{v}} f(\mathbf{a})$ for each unit vector below:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \mathbf{v}_3 = \frac{1}{5}\begin{bmatrix} 3 \\ 4 \end{bmatrix}, \qquad \mathbf{v}_4 = -\frac{1}{5}\begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

Which listed direction gives the greatest increase? Which gives the greatest decrease?

Activity: Directional derivatives and unit directions (U3-LO8)

Let $f(x, y) = x^2y + y^2$.
Compute the directional derivative $D_{\mathbf{v}} f(1, 2)$, where $\mathbf{v} = (1/\sqrt{2}, 1/\sqrt{2})$, and $f(x, y) = x^2y + y^2$.

Activity: Directional derivatives and unit directions (U3-LO8)

Let $f(x, y) = x^2y + y^2$.
Find the directional derivative of f at the point $(-1, 3)$ in the direction of the vector $(-1, \sqrt{3})$. That is, compute $D_{\mathbf{v}}f(-1, 3)$, where $\mathbf{v}$ is the unit vector in the direction of $(-1, \sqrt{3})$.

Activity: A directional derivative between two points (U3-LO8)

Find the directional derivative of f at $(2, 1)$ in the direction from $(2, 1)$ to $(3, 5)$.

Activity: Directional change toward a specified point (U3-LO8)

Let $f(x, y) = x^2 e^y$. Find the directional derivative of f at $(2, 0)$ in the direction pointing from $(2, 0)$ to $(3, 0)$.

Theorem

Suppose that a function f is differentiable at a point $\mathbf{a}$.

- If $\nabla f(\mathbf{a}) = 0$, then $D_{\mathbf{v}} f(\mathbf{a}) = 0$ for any unit vector $\mathbf{v}$.
- If $\nabla f(\mathbf{a}) \neq \mathbf{0}$, then $D_{\mathbf{v}} f(\mathbf{a})$ is maximized, among unit vectors $\mathbf{v}$, when

$$\mathbf{v} = \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}.$$

It is minimized when

$$\mathbf{v} = -\frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}.$$

The maximum value is $\|\nabla f(\mathbf{a})\|$, and the minimum value is $-\|\nabla f(\mathbf{a})\|$.

Why is this true?

If $\nabla f(\mathbf{a}) = \mathbf{0}$, then the theorem gives

$$D_{\mathbf{v}} f(\mathbf{a}) = 0$$

for every unit vector $\mathbf{v}$.

Now suppose $\nabla f(\mathbf{a}) \neq \mathbf{0}$, and let θ be the angle between $\nabla f(\mathbf{a})$ and a unit vector $\mathbf{v}$. By the theorem and Angle between vectors,

$$D_{\mathbf{v}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v} = \|\nabla f(\mathbf{a})\| \cos\theta.$$

The largest possible value of $\cos\theta$ is 1, attained when $\mathbf{v}$ points in the direction of the gradient. The smallest possible value is -1, attained when $\mathbf{v}$ points in the opposite direction. Therefore the maximum and minimum directional derivatives are

$$\|\nabla f(\mathbf{a})\| \quad \text{and} \quad -\|\nabla f(\mathbf{a})\|,$$

attained at

$$\mathbf{v} = \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|} \quad \text{and} \quad \mathbf{v} = -\frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|},$$

respectively.

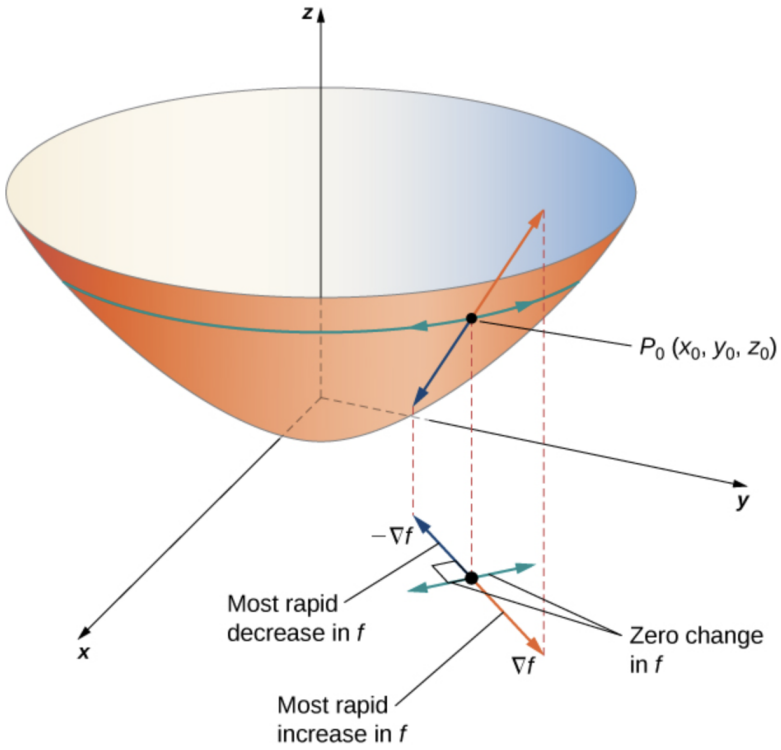


Figure. Gradient directions at a point on a surface.

Activity: Finding the direction of greatest increase (U3-LO8)

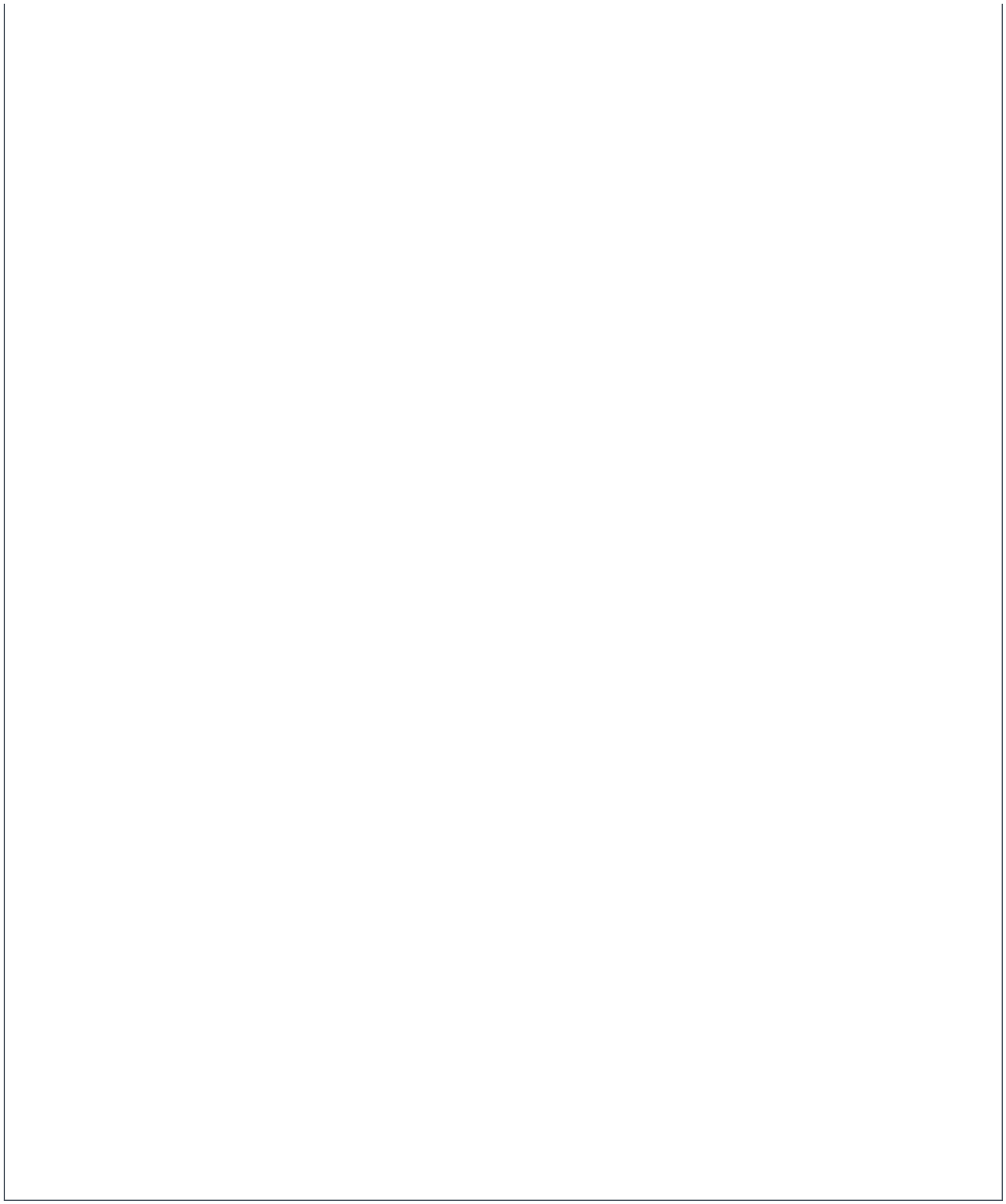
Find the direction for which the directional derivative of $f(x, y) = 3x^2 - 4xy + 2y^2$ at $(-2, 3)$ is a maximum. What is the maximum value of the directional derivative?

Activity: A directional derivative in three variables (U3-LO8)

Calculate $D_{\mathbf{u}} f(1, -2, 3)$ in the direction of $\mathbf{v} = -\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ for the function

Activity: A directional derivative in three variables (U3-LO8) (continued)

$$f(x, y, z) = 5x^2 - 2xy + y^2 - 4yz + z^2 + 3xz.$$



In Orthogonal projection, the same dot product will be interpreted as a signed projection coefficient of the gradient onto the chosen direction.

Optimization problems and critical points

Directional derivatives describe first-order change along a chosen direction. We now use them to identify candidates for local extrema and to motivate an iterative method for decreasing a function.
In this unit, we apply multivariate calculus to solve *optimization problems*.

Definition: Optimization problems

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a scalar-valued function, and consider a set $U \subseteq \mathbb{R}^n$. Then we can consider two optimization problems, in which we call f the objective function of the problem, and U the feasible region of the optimization problem:

- The minimization problem of f on U is the problem of finding $\mathbf{x}^* \in U$ such that the value $f(\mathbf{x}^*)$ is *smallest* among all elements of U (so the function is *minimized* on U by $\mathbf{x}^*$).
- The maximization problem of f on U is the problem of finding $\mathbf{x}^* \in U$ such that the value $f(\mathbf{x}^*)$ is *largest* among all elements of U (so the function is *maximized* on U by $\mathbf{x}^*$).

Definition: Minima and maxima

Suppose f is a scalar-valued function. A point $\mathbf{a}$ in the domain of f is a

- local minimum if there is a ball centered at $\mathbf{a}$ such that $f(\mathbf{a}) \leq f(\mathbf{x})$ whenever $\mathbf{x}$ is in that ball and also in the domain of f; in other words, whenever $\mathbf{x}$ is close enough to $\mathbf{a}$ and lies in the domain of f. In this case, we say $f(\mathbf{a})$ is a local minimum value for the function f.

- local maximum if there is a ball centered at $\mathbf{a}$ such that $f(\mathbf{a}) \geq f(\mathbf{x})$ whenever $\mathbf{x}$ is in that ball and also in the domain of f; in other words, whenever $\mathbf{x}$ is close enough to $\mathbf{a}$ and lies in the domain of f. In this case, we say $f(\mathbf{a})$ is a local maximum value for the function f.
- global minimum if $f(\mathbf{a}) \leq f(\mathbf{x})$ for all $\mathbf{x}$ in the domain of f. In this case, $f(\mathbf{a})$ is the global minimum value, or absolute minimum value, of the function f.
- global maximum if $f(\mathbf{a}) \geq f(\mathbf{x})$ for all $\mathbf{x}$ in the domain of f. In this case, $f(\mathbf{a})$ is the global maximum value, or absolute maximum value, of the function f.

A local minimum or local maximum is also called a local extremum.

Remark

The graph of the function $-f$ is obtained by reflecting the graph of the function f in the output axis. This operation turns local and global maxima into local and global minima, and vice versa. So methods for finding minima for a general function often give methods for finding maxima, which makes the two problems often equivalent.

Definition: Critical points

Suppose f is a scalar-valued function that is differentiable at a point $\mathbf{a}$.

Then we say $\mathbf{a}$ is a **critical point** if $\nabla f(\mathbf{a}) = \mathbf{0}$.

For a scalar-valued function of two variables, for example, $f(x, y)$, a critical point is a solution to the two equations $f_x(x, y) = 0$ and $f_y(x, y) = 0$. Just as the derivative of a scalar-valued function of a single variable helps identify local minima and maxima, partial derivatives provide information about possible local minima/maxima.

Theorem

Suppose f is differentiable at an interior point $\mathbf{a}$ of its domain. If $\mathbf{a}$ is a local extremum of f, then

$$\nabla f(\mathbf{a}) = \mathbf{0}.$$

Why is this true?

Let $\mathbf{v}$ be a unit vector, and define

$$g(t) = f(\mathbf{a} + t\mathbf{v}).$$

Because $\mathbf{a}$ is an interior local extremum of f, the function g has a local extremum at $t = 0$. Therefore

$$g'(0) = 0.$$

By the theorem,

$$g'(0) = D_{\mathbf{v}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v}.$$

This holds for every unit vector $\mathbf{v}$. If $\nabla f(\mathbf{a}) \neq \mathbf{0}$, choose

$$\mathbf{v} = \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}.$$

Then

$$\nabla f(\mathbf{a}) \cdot \mathbf{v} = \|\nabla f(\mathbf{a})\| > 0,$$

a contradiction. Hence $\nabla f(\mathbf{a}) = \mathbf{0}$.

Activity: Critical points of a quadratic function (U3-LO9)

Find the critical points of the function

$$f(x, y) = x^2 + y^2 - 2x + 4y.$$

Activity: Critical points of a difference of squares (U3-LO9)

Find the critical points of $f(x, y) = x^2 - y^2$.

Warning: Critical points are candidates

The equation $\nabla f = 0$ finds candidates for local extrema. It does not by itself find boundary extrema, constrained extrema, or absolute extrema. Always combine critical-point information with the domain and boundary conditions.

Gradient descent

Activity: Critical-point equations in three variables (U3-LO9)

Find the system of equations satisfied by the critical points of the function

$$f(x, y, z) = x^2 e^{yz} + y \sin(xz) + z \cos(xy).$$

Using the calculus we have developed, it is usually straightforward to write the equations that critical points must satisfy. Critical-point equations in three variables (U3-LO9) also shows the problem: those equations may be nonlinear and difficult or impossible to solve exactly. This is different from the linear systems in Unit 2, where Gaussian elimination gave a systematic exact method.

When the critical-point equations cannot be solved directly, one often uses a numerical method. By the theorem, the negative gradient gives the direction of steepest local decrease.

At a current point $\mathbf{x}_k$, choose

$$\mathbf{h} = -\alpha \nabla f(\mathbf{x}_k), \qquad \alpha > 0.$$

By Linear Approximation,

$$\begin{aligned} f(\mathbf{x}_k + \mathbf{h}) - f(\mathbf{x}_k) &\approx \nabla f(\mathbf{x}_k)^T \mathbf{h} \\ &= -\alpha \|\nabla f(\mathbf{x}_k)\|^2. \end{aligned}$$

Unless the gradient is zero, the local model predicts a decrease. This is a local prediction, not a convergence theorem. It motivates the gradient-descent update

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k).$$

The gradient-descent algorithm repeats this local step.

1. Start at an initial point $\mathbf{x}_0$.
2. Compute the gradient $\nabla f(\mathbf{x}_0)$.
3. Move in the direction of the negative gradient: set $\mathbf{x}_1 = \mathbf{x}_0 - \alpha_0 \nabla f(\mathbf{x}_0)$, where $\alpha_0 > 0$ is a step size (or learning rate) parameter.
4. Repeat steps 2-3 with step sizes $\alpha_1, \alpha_2, \ldots$ to successively define points $\mathbf{x}_2 = \mathbf{x}_1 - \alpha_1 \nabla f(\mathbf{x}_1)$, $\mathbf{x}_3 = \mathbf{x}_2 - \alpha_2 \nabla f(\mathbf{x}_2)$, and so on, with the hope that these points converge to a local minimum.

In data and modeling problems, an objective function being minimized is often called a loss.
Intuitively, gradient descent works because each successive value $\mathbf{x}_k$ is obtained by shifting the previous value $\mathbf{x}_{k-1}$ in the direction in which the function f decreases the fastest. Gradient descent may converge to a local minimum under additional assumptions on the function and the learning rates. The examples below illustrate possible behavior; they do not prove convergence.

Activity: Reading one gradient descent update (U3-LO8, U3-LO9)

Assume `grad_f(x)` computes $\nabla f(\mathbf{x})$. Consider the line of code

```
x = x - alpha * grad_f(x)
```

1. What mathematical update rule does this line represent?
2. Which quantity is the learning rate?
3. Why must `grad_f(x)` have the same shape as `x`?
4. What would change if the minus sign were a plus sign?

Activity: Three gradient-descent steps (U3-LO9)

Find an approximate minimum of the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) = x^2 - 3xy + 3y^2 + 5y + 2x$$

using gradient descent. Use the starting point $\mathbf{x}_0 = (0, 0)$, a constant step size of 0.1, and stop after computing $\mathbf{x}_3$. You may use a calculator to perform these computations.

Activity: From a symbolic gradient to a descent step (U3-LO9)

Use SymPy to check the first step of the preceding calculation for $f(x, y) = x^2 - 3xy + 3y^2 + 5y + 2x$. We construct a gradient as in Building a gradient and Hessian in SymPy (U3-LO3), evaluate it at the starting point, and use the vector update from Reading one gradient descent update (U3-LO8, U3-LO9). The learning rate `sp.Rational(1, 10)` represents 0.1 exactly.

```
import sympy as sp

x, y = sp.symbols("x y")
f = x**2 - 3*x*y + 3*y**2 + 5*y + 2*x
grad_f = sp.Matrix([sp.diff(f, x), sp.diff(f, y)])

p0 = sp.Matrix([0, 0])
```

Activity: From a symbolic gradient to a descent step (U3-LO9) (continued)

```
g0 = grad_f.subs({x: p0[0], y: p0[1]})
alpha = sp.Rational(1, 10)
p1 = p0 - alpha*g0
f0 = f.subs({x: p0[0], y: p0[1]})
f1 = f.subs({x: p1[0], y: p1[1]})

print(grad_f)
print(g0)
print(p1)
print(f0, f1)
print((g0.T * (p1 - p0))[0], f1 - f0)
```

Output:

```
Matrix([[2*x - 3*y + 2], [-3*x + 6*y + 5]])
Matrix([[2], [5]])
Matrix([[-1/5], [-1/2]])
0 -241/100
-29/10 -241/100
```

In the last line, `g0.T * (p1 - p0)` is a 1×1 matrix. The index `[0]` extracts its scalar entry, the local linear prediction for the change in f.

1. What is the difference between `grad_f` and `g0`? What shapes must `p0`, `g0`, and `p1` have for the update to make sense?
2. Write the numerical vector update that produces `p1`. Did the function value decrease? Compare the predicted change with the actual change, using the scalar linear approximation from Linear Approximation.
3. Predict `grad_f.subs({x: p1[0], y: p1[1]})`. Is `p1` a critical point? Explain which symbolic expression can be reused at the next step and which evaluated vector must be recomputed.

Activity: From a symbolic gradient to a descent step (U3-LO9) (continued)

4. If `alpha` is changed to `1` and the subsequent lines are rerun, what are `p1` and `f1`? What does this comparison show about the learning rate?

Remark

The iteration is cumbersome by hand but easy to repeat on a computer when the gradient is available.

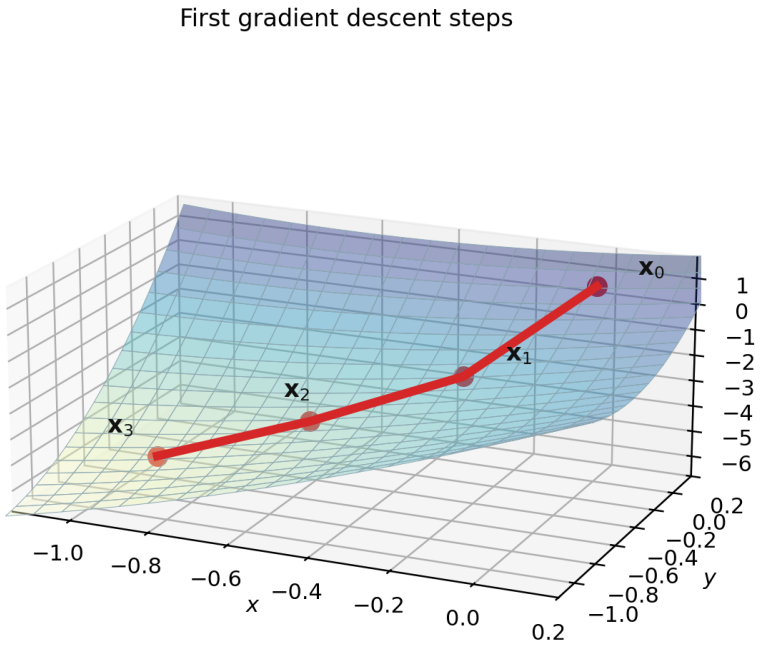


Figure. Surface plot showing the first gradient descent steps from the origin.

Activity: Improving a model's predictions with one step (U3-LO8, U3-LO9)

Return to the three observations from From predictions to a scalar error function (U3-LO1, U2-LO1) and the loss

$$\mathscr{L}(w_1, w_2) = (w_1 - 1)^2 + (w_2 - 2)^2 + (w_1 + w_2 - 2)^2.$$

At the starting weights $\mathbf{w}_0 = (1, 1)$, the loss is 1 and the gradient is

Activity: Improving a model's predictions with one step (U3-LO8, U3-LO9) (continued)

$(0, -2)$. In this model, a training step adjusts the weights while the features X and observations $\mathbf{y}$ remain fixed.

1. Find the unit direction of greatest decrease at $\mathbf{w}_0$ and its directional derivative. Take one gradient-descent step with learning rate $\alpha = 1/10$.
2. Compute the new prediction vector, residual, and loss. Compare the actual change with the first-order prediction $-\alpha\|\nabla\mathscr{L}(\mathbf{w}_0)\|^2$.
3. Evaluate the gradient at the new weights. Has the step reached a critical point?
4. Starting again at $\mathbf{w}_0$, what happens if $\alpha = 1$? Explain why the same descent direction can give an increase in loss.

Activity: Redundant features give a flat direction of the loss (U3-LO6, U2-LO6)

Recall Unit 2's redundant-feature example in Two coefficient choices, same predictions (U2-LO1, U2-LO6). Add a third feature to our data by summing the original two feature columns:

$$\widetilde{X} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

There are now three weights $\theta = (\theta_1, \theta_2, \theta_3)$, with loss

$$\widetilde{\mathscr{L}}(\theta) = \|\widetilde{X}\theta - \mathbf{y}\|^2.$$

The predictions depend only on the two combinations $w_1 = \theta_1 + \theta_3$ and $w_2 = \theta_2 + \theta_3$, so $\widetilde{\mathscr{L}}(\theta) = \mathscr{L}(\theta_1 + \theta_3, \theta_2 + \theta_3)$.

Activity: Redundant features give a flat direction of the loss (U3-LO6, U2-LO6) (continued)

1. Verify that $\mathbf{z} = (-1, -1, 1)$ is in the null space of $\widetilde{X}$.
2. Show that $\widetilde{\mathscr{L}}(\theta + t\mathbf{z}) = \widetilde{\mathscr{L}}(\theta)$ for every real t. Evaluate this constant loss along the line through $\theta_0 = (1, 1, 0)$.
3. What is $\frac{d}{dt}\widetilde{\mathscr{L}}(\theta + t\mathbf{z})$? If a minimizing weight vector is found, can it be the unique minimizing weight vector? Compare this exact invariance with the local Jacobian example in Interpreting a Jacobian's null space in SymPy (U3-LO5, U2-LO3).

These examples use derivatives to improve a fit and interpret directions that leave it unchanged. Unit 4 develops the general least-squares method for finding a best-fitting prediction vector and its geometric interpretation.